WEEKLY INTERNSHIP ACTIVITIES & ASSIGNMENT 4

ACTIVITY:

Title: Date-Time Data Processing and Analysis Exercise

Description: Students will work with datasets containing date-time fields to practice parsing, formatting, and extracting components such as year, month, day, and weekday. The exercise emphasises understanding why temporal data is critical in analytics workflows.

Reference/Source Code/Links:

- https://pandas.pydata.org/docs/user_guide/timeseries.html
- https://strftime.org/

Objectives:

- Understand importance of date-time data in analytics
- Practice parsing and formatting date values
- Extract useful time components
- Perform basic time-based calculations such as durations

Deliverables

- Script demonstrating date-time processing
- Output table showing extracted components
- Short explanation (400–600 words)

ASSIGNMENT:

Title: Simple Linear Regression Modeling Project

Description: Students will implement a basic linear regression model, interpret the regression equation, and discuss assumptions such as linearity and independence.

Objectives

- Understand intuition behind linear regression
- Learn regression equation structure
- Recognise assumptions of linear regression
- Practice fitting a simple model conceptually or through code

Reference/Source Code/Links: Branding strategy articles:

- https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegre ssion.html
- https://developers.google.com/machine-learning/crash-course/linear-regression

Deliverable:

- Regression model script or notebook
- Interpretation summary (600–800 words)
- Regression plot visualisation